

Education

Iowa State University

Master of Science in Mechanical Engineering

GPA : 3.91 / 4.0

Aug 2024 – May 2026 (Expected)

Ames, Iowa

Visvesvaraya Technological University

Bachelor of Mechanical Engineering

GPA : 3.61 / 4.0

Aug 2019 – Jul 2023

India

Relevant Coursework

- Linear Systems
- Electronic System Integration & Control
- Big Data Analytics & Optimization
- Machine learning for Cyber-Physical Systems
- Solid Modeling & GPU Computing
- Optimization for Complex Design

Research Experience

Research Assistant - NSF Funded AI-Based AFM Project

May 2025 – Present

Iowa State University

Ames, Iowa

- Contributing to an NSF-funded project on AFM nanopositioners, capturing **system behavior** and improving precision through **advanced control and ML methods**.
- Designed a deep learning-based **sequence-to-sequence LSTM framework** for modeling of nonlinear actuator dynamics.
- Developed and implemented **black-box identification** pipelines for nonlinear hysteretic systems.
- Built experimental acquisition workflows for time-series signals and conducted **time and frequency-domain validation**.
- Designed and validated **ML-based control algorithms** for nonlinear, **noisy system dynamics**.
- **Reduced training time by 50%** using GPU-based computation and **B-spline decomposition** for feature compression.

Teaching Assistant: Heat Transfer(ME 4360), System Dynamics & Control (ME 4210)

Aug 2024 – Present

Iowa State University

Ames, Iowa

- Supported **100+ students** through labs, recitations, and grading, applying statistical methods to assess performance trends and improve instructional outcomes.
- Improved student learning outcomes, reflected in strong performance feedback (**4.3/5.0**).
- Facilitated lab safety protocols and trained students on specialized thermal equipment and sensors.

Research Intern – Nanometrology Lab

Jul 2023 – Feb 2024

Indian Institute of Science (IISc)

Bangalore, India

- Investigated and refined **micro-robotic compliant mechanism** designs for bridge-amplifier nanopositioners used in in-house AFM systems, contributing to improved precision and imaging stability.
- Conducted **static and modal characterization** of compliant mechanisms using **COMSOL Multiphysics** and **MATLAB** to evaluate performance and guide design decisions.
- Supported prototype development by collaborating with fabrication teams using micro-fabrication methods.

Notable Projects

Identification of nonlinear system behavior through ML-based control and B-spline feature encoding

- Developed a framework to learn nonlinear actuator mappings via B-spline representation and LSTM training.
- Integrated spline-based feature extraction for interpretability in time-domain system identification.

Road Segmentation for Autonomous Vehicles | *Computer Vision Coursework project*

- Implemented Fast-SCNN, SegFormer, YOLOv8, and SAM for semantic & instance segmentation.
- Trained on CityScapes dataset using PyTorch, OpenCV.

Self Stabilizing spoon (Assistive device) | *Bachelors major project*

- Implemented real-time motion control using MEMS gyro and servo feedback loops (PID).
- Tuned gains experimentally; identified plant dynamics using frequency-sweep data.
- Demonstrated application of system identification for vibration compensation.

Technical Skills

Languages: Matlab, Python, C

Tools: Simulink, COMSOL Multiphysics, Ansys, CANoe, Nanoscope

Frameworks: Pytorch, GitHub, Matlab Deep-learning toolkit

Leadership

SAE Collegiate Design Series - East

March 2023

Team Captain

Lakeland, Florida

- Project selected at **International level** and was invited to participate at SAE Aero Design Series competition held at **Lakeland, Florida USA**.
- **Captained** a team of 18 members and achieved **6th place finish out of 20 teams** from **reputed universities worldwide**.
- Designed an aircraft that was capable of being airborne within **9' of take off** with less than **450W** of power and equipped with a fail safe mechanism.